

Barcelona also has improved data fetch bandwidth and branch prediction, and AMD borrows Out of Order Execution from Intel (Core 2). All in all, in Barcelona AMD sees deliverance from the thorn that Core 2 has become. The Barcelona architecture will also make its presence felt in the notebook arena eventually replacing the current Turion X2 processors. In Desktop avatar, Barcelona has been codenamed "Phenom."

The Cell processor powering the PS3 should, we reckon, head for computers soon. Designed ground up for high-performance computing the Cell processor consists of data and program cells (software) that are the fodder for the hardware cells. The Cell architecture isn't restricted to computing on a single device, and if you have two or more Cell powered devices hooked up they can easily cooperate while processing. This is called "Distributed Computing". Scalability is the name of the game with the Cell architecture that is

Manufacturers will have the option to simply plonk different cores onto a single wafer, say three graphics cores and one general processing core—based on demand

designed for use in Desktops, Servers, Cell Phones, PDAs and even Consoles, so software cells can be distributed to each of these devices.

The performance of a single Cell processor is 256 billion floating point operations per second (256 GFLOPS). In comparison a Core 2 Duo e6600 at 2.4 GHz manages just around 4 GFLOPS!

We could well see your cell phone assist your console while gaming or your PC chipping in to share the load with your PDA. Convergence? Or parallelism unparalleled? Applications are pushing the bounds of interactivity and realism, and this bridge may only be crossed through faster processing. A small example is Windows Vista—a gorgeous bit of software that, while being sinfully delightful to use, exacts a heavy toll on hardware. And if better computing devices are available for the same (if not reduced) prices, what's the problem? "May you live in interesting times"... was that the way the Chinese proverb went? Wish granted! ☒

Graphics Cards

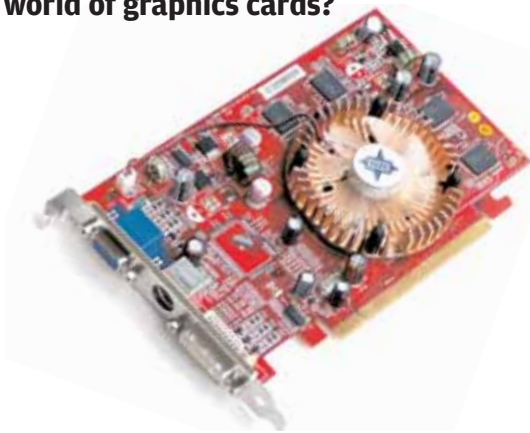
What's up next in the blink-and-miss-it world of graphics cards?

The only category of computing hardware more exciting than processors at the moment—or for that matter in an even greater state of flux—is the GPU. And it's infinitely more noticeable too. Seeing is equal to believing and GPUs are all about the visual experience.

Graphics cards have lost the stigma of being a "gamer's tool." We've seen greater acceptance of these processing marvels in many a home. The reason is the burgeoning drive for all things multimedia—movies, digital imaging, and, yes, games. And it's the entertainment industry by large that drives GPU manufacturers back to their drawing boards to come out with solutions capable of driving their products. And the entertainment industry isn't self-driven either—it's driven by us! It's in our nature to notice and applaud beauty. A simple example—anyone who has used Windows XP for a few days will find moving back to Windows 98 painful, inconvenient, and messy not to mention an eyesore! Take a wild guess as to what would happen if this new Windows XP fan were to be exposed to Windows Vista for a few days?

As with CPUs, the world of add-on desktop / notebook graphics solutions is a two brand show—NVIDIA and ATI. The phenomenon of one-up is seen at work here. NVIDIA had some success with their move to a 90 nm fabrication process with their G71 (7900 series) and G73 (7600 series) parts, which were adopted by gamers and multimedia aficionados alike. We've seen what ATI is capable of, courtesy the 48-pixel units on their X1900XTX series. These cards were DirectX 9.0c ready, and played most (then) current-generation games with a blatant absence of stuttering frames!

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designing a solution for an API that hasn't seen the light of day... yet. This API would obviously need a lot more processing power, and current DX 9 games were pushing the latest cards anyway.

Out came unified shaders (no more separate pixel and vertex shaders)—and NVIDIA beat ATI to the launch with 128 such shaders onboard their 8800GTX (codenamed G80). This 681-million transistor behemoth features, for the first time, a 384-bit memory interface, which in itself provides a huge theoretical advantage over the previously de facto 256 bits. Lower variants of the GTX designated "GTS" followed.

NVIDIA has since debuted the mid-range solution of their 8-series, a.k.a. the 8600GTS and 8600GT, which also signify the move to an 80 nm manufacturing process. Featuring a much lower shader unit count and slower and less memory, these have been very favourably priced, and we've seen these cards sell on the street for below Rs 10,000 (GT), while the GTS costs around Rs 3,000-4,000 more. We expect these prices to fall further as other DX 10 cards release and the dawn of the API gets nearer.

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